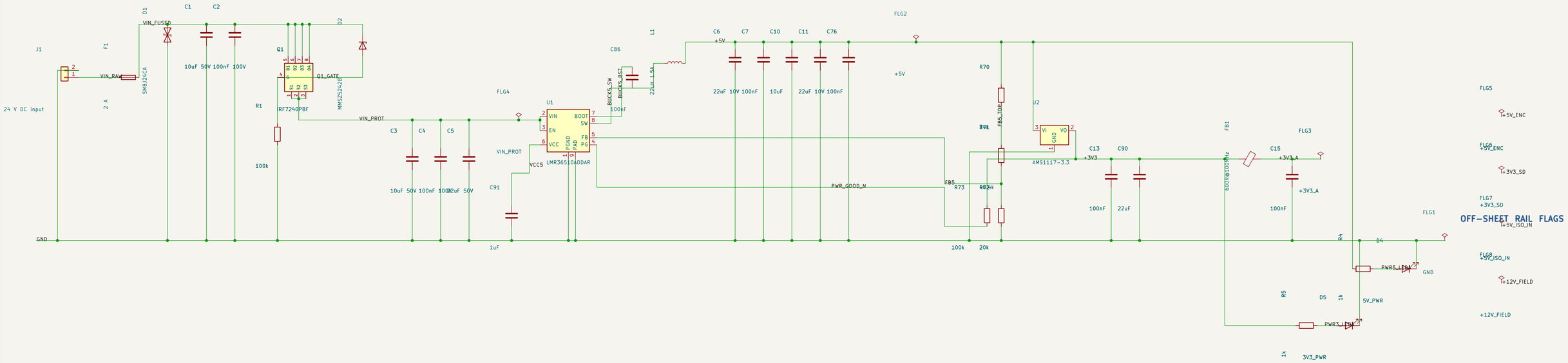
Power flow: 24 V input -> fused/protected VIN -> 5 V buck -> 3V3 LDO

INPUT • FUSE • REVERSE-POLARITY PROTECTION

5 V BUCK POWER STAGE

3V3 LDO • FILTER • STATUS

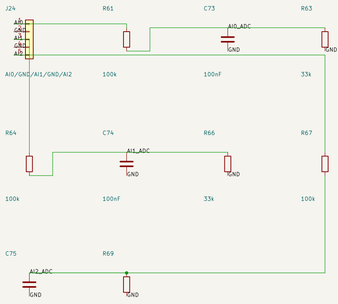


OFF-SHEET RAIL FLAGS

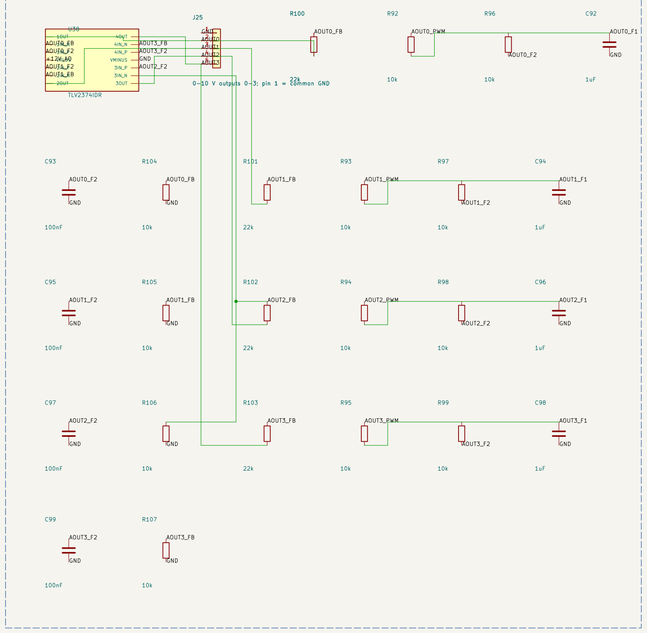
Method B: existing symbols, explicit orthogonal wiring	
LeanCam / uCNC experiment	
Sheet: /	
File: lean-power-b.kicad_sch	
Title: LeanCam NCC RP2350B – Power and Supervision (Python re-layo	
Size: A2	Date: 2026-10-29
KiCad E.D.A. 10.0.1	**Rev:** Id: 1/1

Analog inputs and filtered 0–10 V outputs | 40 components | 29 local nets

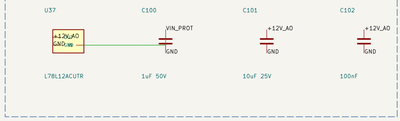
ANALOG INPUTS AI0–AI2



ANALOG OUTPUTS AO0–AO3



ANALOG 12 V SUPPLY



Existing symbols grouped by function; local nets routed where practical

LeanCam / uCNC Python hierarchy experiment

Sheet: /

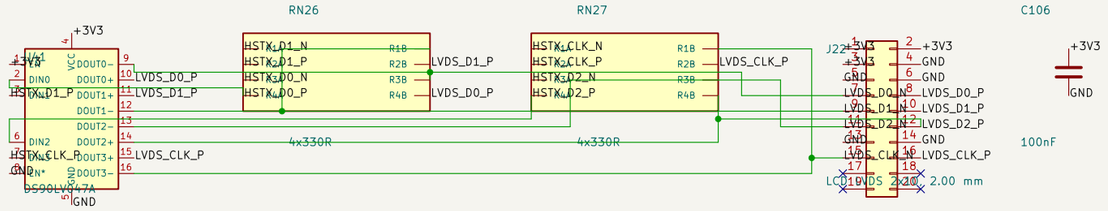
File: analog_io_python.kicad_sch

Title: Analog inputs and filtered 0–10 V outputs

Size: A1	Date: 2026-10-29	**Rev:**
KiCad E.D.A. 10.0.1		id: 1/1

LVDS display interface | 5 components | 18 local nets

LVDS TRANSMITTER AND CONNECTOR



Existing symbols grouped by function; local nets routed where practical

LeanCam / uCNC Python hierarchy experiment

Sheet: /

File: display-python.kicad_sch

Title: LVDS display interface

Size: A3	Date: 2026-10-29	**Rev:**
KiCad E.D.A. 10.0.1		Id: 1/1

Encoder receiver and 5 V protection | 9 components | 22 local nets

ENCODER INPUTS 0–2

F2
+5V
+5V_ENC
500mA

D53
+5V_ENC
GND
SMAJ5.0A

C57
+3V3
GND
100nF

U28
SN74AHCT541DW
GND
OE1_N
A0
Y0
A1
Y1
ENC0_B_5V
ENC0_Z_5V
A3
Y3
GND
VCC
OE2_N
Y7
A7
Y6
A6
Y5
A5
Y4
A4
+3V3
GND
ENC0_A_3V3
ENC0_B_3V3
ENC0_Z_3V3
ENC1_A_3V3
ENC1_B_3V3
ENC1_Z_3V3
ENC2_A_3V3
ENC2_B_3V3

J13
+5V_ENC
ENC0_A_5V
ENC0_B_5V
ENC0_Z_5V
GND
Encoder 0: +5,A,B,Z,GND

J14
+5V_ENC
ENC1_A_5V
ENC1_B_5V
ENC1_Z_5V
GND
Encoder 1: +5,A,B,Z,GND

J15
+5V_ENC
ENC2_A_5V
ENC2_B_5V
ENC2_Z_5V
GND
Encoder 2: +5,A,B,Z,GND

R110
ENC2_Z_3V3
5.1k

R111
GND
10k

Existing symbols grouped by function; local nets routed where practical

LeanCam / uCNC Python hierarchy experiment

Sheet: /
File: encoders-python.kicad_sch

Title: Encoder receiver and 5 V protection

Size: A3	Date: 2026-10-29	**Rev:**
KiCad E.D.A. 10.0.1		Id: 1/1

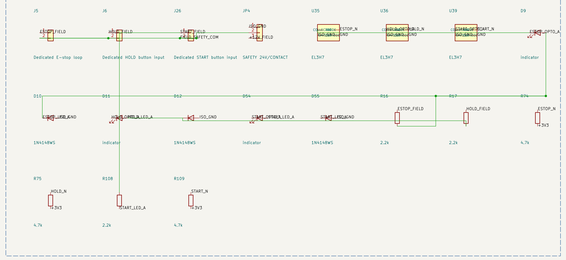
ISOLATED 5 V-TO-12 V SUPPLY



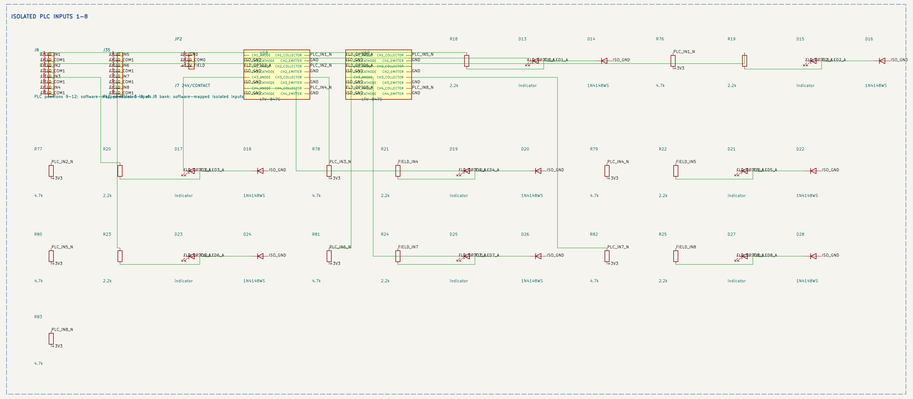
EXPANSION INPUTS 0–7



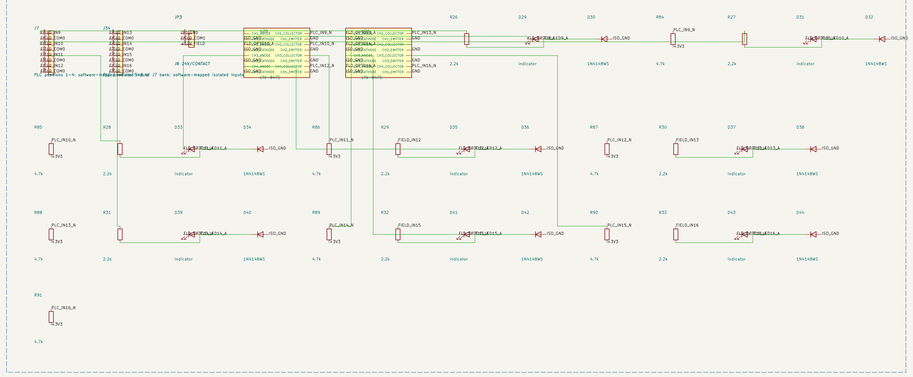
SAFETY INPUTS



ISOLATED PLC INPUTS 1–8

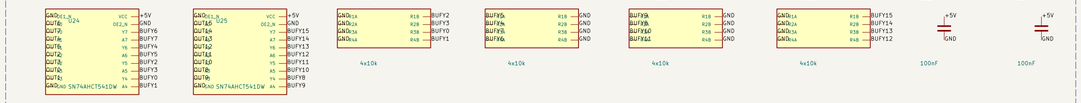


ISOLATED PLC INPUTS 9–16

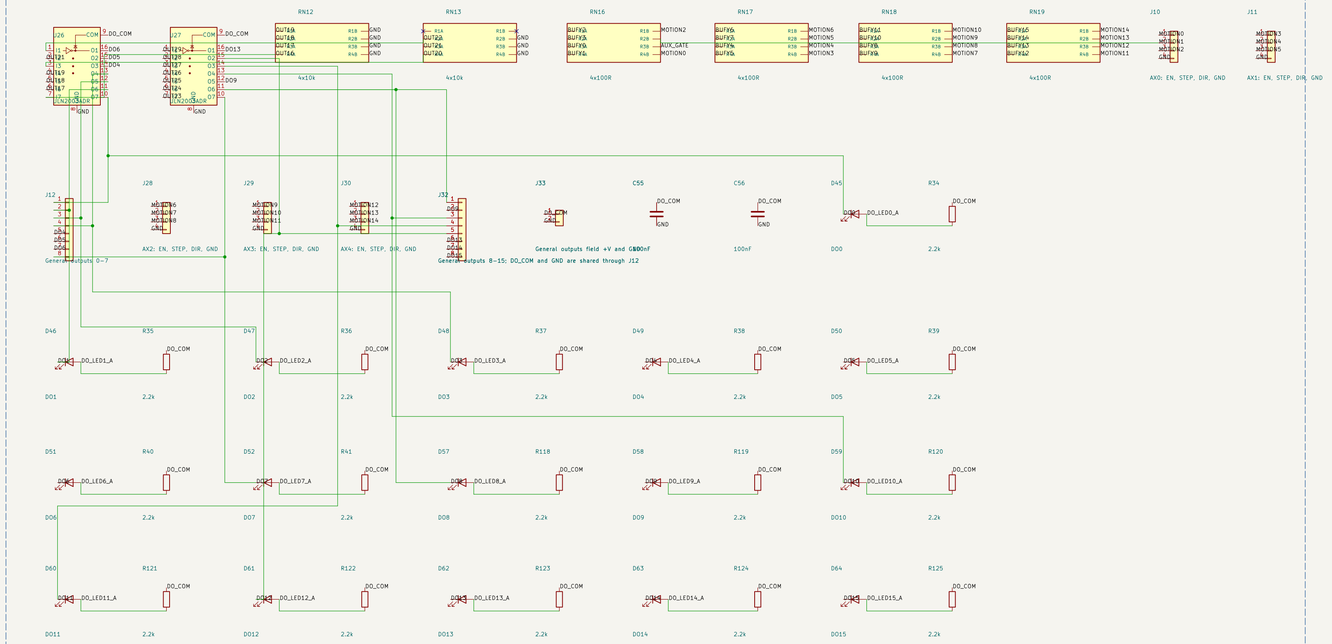


PLC outputs, motion outputs and auxiliary drivers | 71 components | 110 local nets

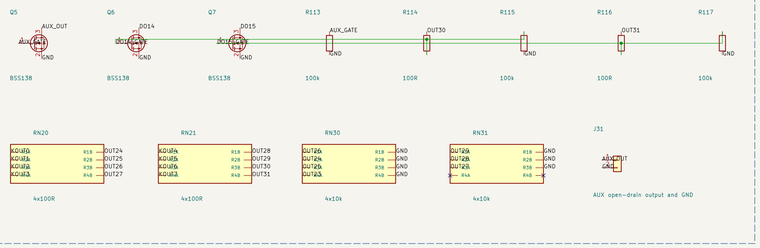
5 V OUTPUT BUFFERS



GENERAL AND MOTION OUTPUTS



AUXILIARY AND HIGH-CURRENT OUTPUTS



Existing symbols grouped by function; local nets routed where practical

LeanCam / uCNC Python hierarchy experiment

Sheet: /

File: field_outputs-python.kicad_sch

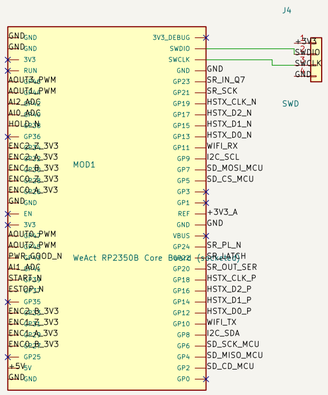
Title: PLC outputs, motion outputs and auxiliary drivers

Size: A1 | Date: 2026-10-29 | **Rev:**

KiCad E.D.A. 10.0.1 | Id: 1/1

RP2350B module, shift-register chains and safety I/O | 23 components | 119 local nets

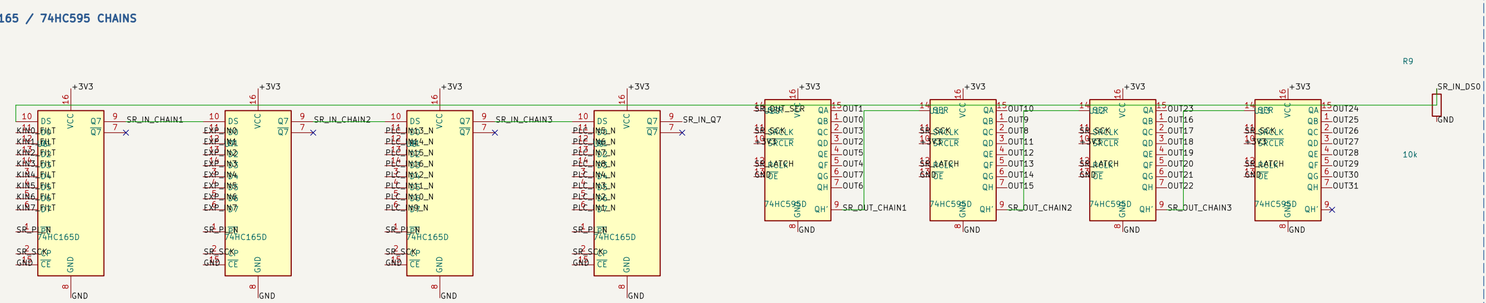
WeAct RP2350B AND DECOUPLING

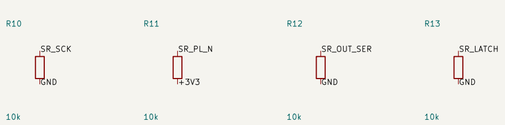






74HC165 / 74HC595 CHAINS





Existing symbols grouped by function; local nets routed where practical

LeanCam / uCNC Python hierarchy experiment

Sheet: /

File: mcu_io-python.kicad_sch

Title: RP2350B module, shift-register chains and safety I/O

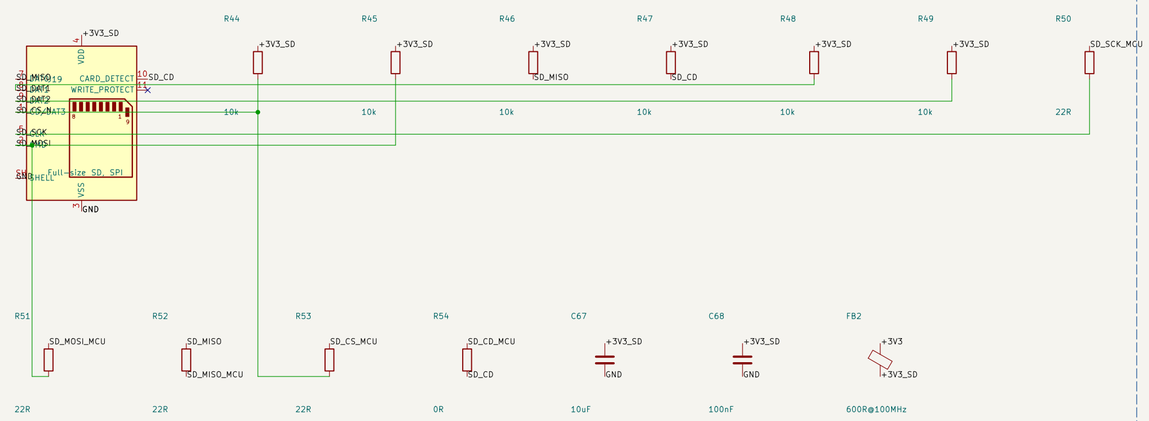
Size: A2	Date: 2026-10-29	**Rev:**
KiCad E.D.A. 10.0.1		Id: 1/1

Storage, Wi-Fi, keyboard and service interfaces | 31 components | 49 local nets

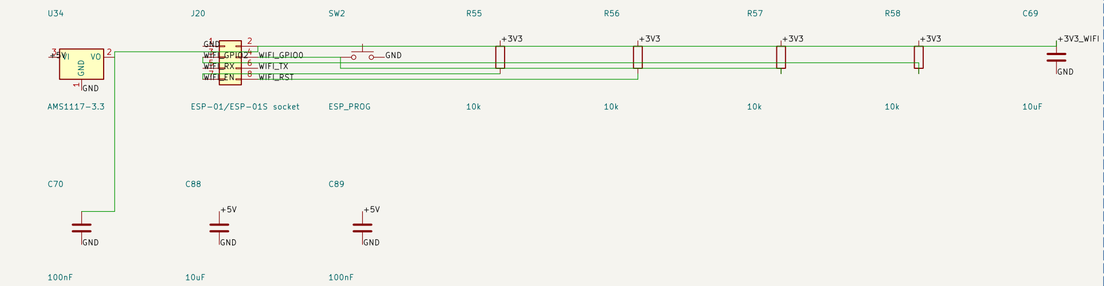
KEYBOARD MATRIX INTERFACE



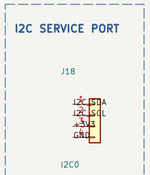
MICRO SD INTERFACE



ESP-01 / ESP-01S SOCKET



I2C SERVICE PORT



Existing symbols grouped by function; local nets routed where practical

LeanCam / uCNC Python hierarchy experiment

Sheet: /
File: storage_wifi_ui-python.kicad_sch

Title: Storage, Wi-Fi, keyboard and service interfaces

Size: A2 | Date: 2026-10-29 | **Rev:**
KiCad E.D.A. 10.0.1 | Id: 1/1